

MGB1121/ CGB1122/EGB1121 – DATA STRUCTURES PROJECT REVIEW-3

Academic Year: 2025 – 2026 (Even Semester)

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Section	: A4
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Project Title

SHORTEST PATH FINDER



ABSTRACT

The Shortest Path Finder project focuses on finding the minimum distance between two vertices in a graph using data structures. In this project, a graph is used to represent nodes and edges, where edges may contain weights indicating distance or cost. The main objective is to determine the shortest and most efficient path from a source node to a destination node.

This problem is solved using graph-based algorithms and data structures such as adjacency lists, arrays, and priority queues. The project helps in understanding how graphs are applied in solving real-world problems like transportation systems, network routing, and navigation. Overall, this project demonstrates the practical use of data structures in optimizing path selection.



ABSTRACT WITH CO & PO MAPPING

Abstract	CO	POs
The Shortest Path Finder project aims to find the minimum distance between two vertices in a graph. A graph is used to represent nodes and weighted edges, where weights indicate distance or cost. The main goal is to determine the most efficient path from a source to a destination. This project uses data structures like adjacency lists, arrays, and priority queues. It shows how graph algorithms are applied in real-world areas such as transportation, network routing, and navigation systems.	CO4 C01	PO1(3) PO2(3) PO3(2) PO5(2)



INTRODUCTION

Shortest Path Finder is a basic concept in Data Structures used to find the minimum distance between two points in a graph. In this project, a graph data structure is used where nodes represent different locations and edges represent the connection between them. Each connection may have a weight that shows the distance or cost.

The main aim of this project is to find the shortest and most efficient path from a starting node to a destination node. This problem is solved using graph-based methods in data structures. It helps in understanding how graphs are used to solve real-world problems like maps and network routing.



DATA STRUCTURES USED

Graph – The main data structure used in this project. It represents vertices as nodes and edges as connections between them. Each edge may contain a weight that indicates distance or cost.

Adjacency List / Adjacency Matrix – These structures are used to store the graph efficiently. They help in representing the relationship between vertices in a clear and organized manner.

Array – Used to store and update the shortest distance values from the source node. It also keeps track of visited vertices during execution.

Priority Queue – Used to select the next vertex with the minimum distance. This helps in improving efficiency while finding the shortest path.



ADVANTAGES

- 1.Helps in finding the shortest and most efficient path between two vertices.
- 2.Reduces time and cost in real-world applications like transportation and network routing.
- 3.Improves understanding of graph data structures and algorithms.
- 4.Provides optimized solutions for complex path problems.
- 5.Can be applied in navigation systems and communication networks.



BLOCK DIAGRAM



BLOCK DIAGRAM - DESCRIPTION

The Shortest Path Finder system is a user-friendly application developed to determine the most efficient route between two selected cities. At the beginning, the user logs into the system by entering a username, which allows access to the main interface. After logging in, the system displays a list of available cities in dropdown menus, from which the user selects a starting point and a destination. Once the cities are chosen, the system processes this input and applies the Dijkstra algorithm, a well-known algorithm used for finding the shortest path in a weighted graph.

The graph in this system represents cities as nodes and the distances between them as edges with weights. The Dijkstra algorithm works by initializing distances and iteratively selecting the nearest unvisited node, updating the shortest distance to its neighboring nodes. Through this process, the algorithm efficiently calculates the minimum distance required to travel from the selected starting city to the destination city. It also keeps track of the path taken to achieve this minimum distance.

After completing the calculation, the system generates the optimal route and displays it clearly to the user along with the total distance in kilometers. In addition to the textual output, the system enhances user understanding by visually representing the cities and connections using a graph. The shortest path is highlighted distinctly in the graph, making it easy for users to identify the route. Overall, this system combines algorithmic efficiency with an interactive interface to provide an effective solution for route optimization and pathfinding problems.



LIST OF MODULES

1.User Login Module – User enters name and logs into the system.

It validates input and provides access to the application.

2.City Selection Module – User selects start and destination cities.

Cities are loaded from the graph and shown in dropdown lists.

3.Graph Module – Stores cities and distances as a graph.

Nodes represent cities and edges represent distances.

4.Shortest Path Module – Uses Dijkstra algorithm to find shortest path.

It calculates minimum distance and tracks the optimal route.

5.Result Module – Displays route and total distance.

Output is shown clearly for easy understanding.

6.Visualization Module – Shows graph with nodes and edges.

Helps users visually understand city connections.

7.Path Highlight Module – Highlights shortest path in the graph.

Uses color (red) to clearly show the optimal route.

8.Admin Module – Shows total users (optional feature).

Allows monitoring of system usage.



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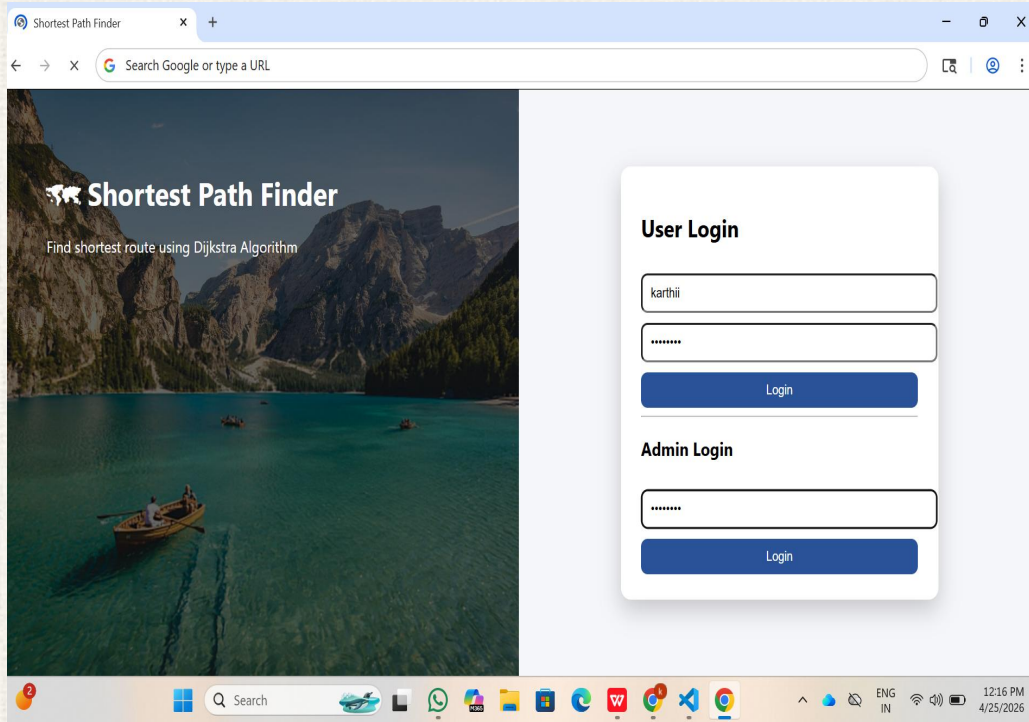
MODULE DESCRIPTION(CONT...)

Each module in the Shortest Path Finder plays an important role in achieving the final result. The input and representation modules ensure that the graph is properly defined and stored. The initialization module sets up the initial conditions required for processing.

The Dijkstra algorithm and distance update modules work together to compute the shortest distances efficiently. The path calculation module helps in identifying the exact route, and the output module presents the final result clearly. This modular design improves the performance and makes the system easy to understand and implement.



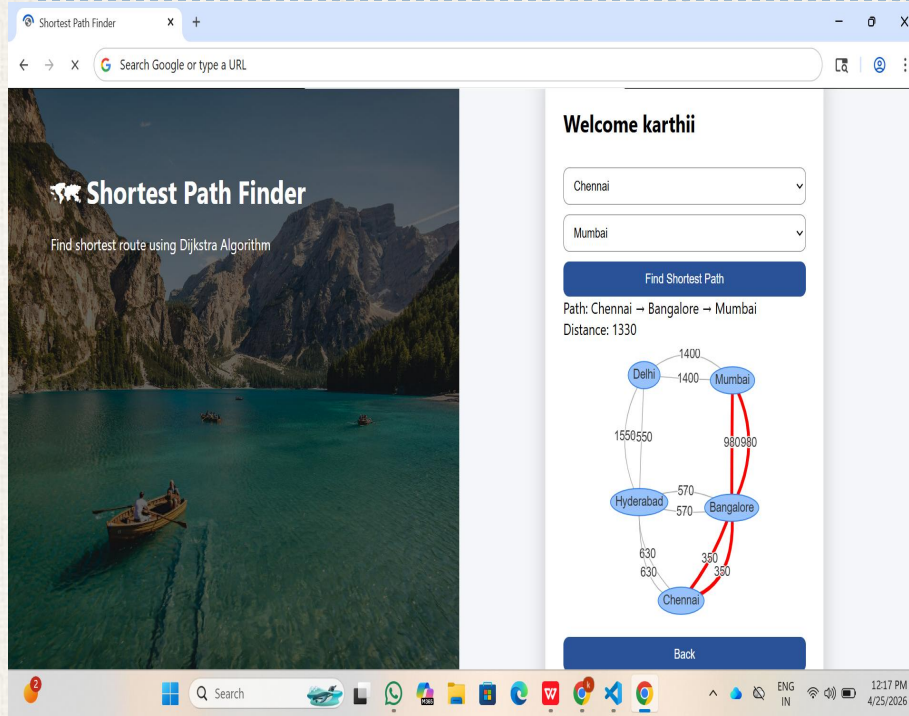
RESULTS AND DISCUSSION



This output shows the login interface of the Shortest Path Finder system. The screen displays a simple and user-friendly design where the user is required to enter a username in the input field.

After entering the name (e.g., “Traveler”), the user can click on the “Login” button to access the system. This step ensures that only authorized users can proceed to use the application features. The clean layout and centered design make the interface easy to understand and use.

RESULTS AND DISCUSSION(CONT...)

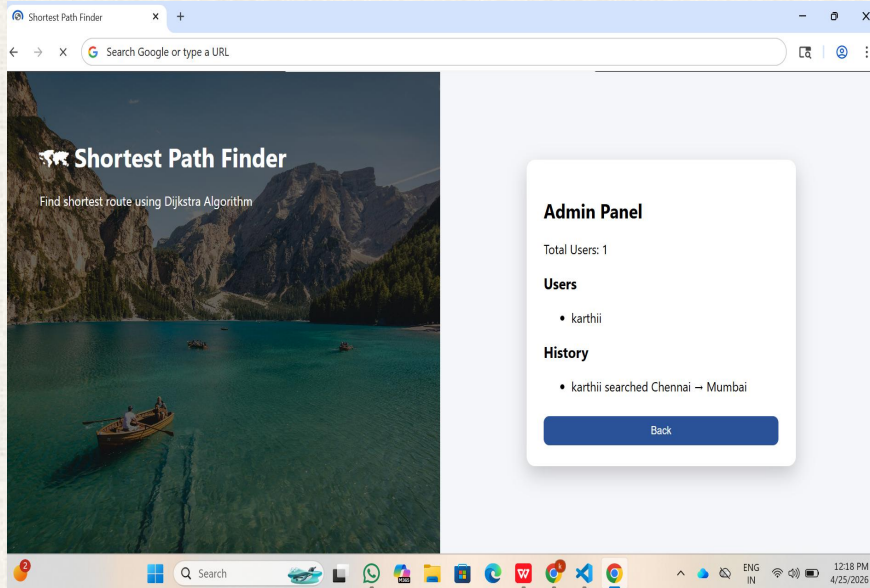


Route Selection and Result Display

This output shows the main interface after user login. The user is welcomed with a message and provided with dropdown menus to select the source and destination cities (e.g., Chennai and Mumbai).

After clicking the “Find Route” button, the system calculates the shortest path and displays the result clearly. The output includes the selected route (Chennai → Bangalore → Mumbai) along with the total distance (1330 km), making it easy for the user to understand the travel path.

RESULTS AND DISCUSSION(CONT...)



This output represents the graphical visualisation of the city network. Cities are shown as nodes, and the connections between them are represented as edges labeled with distances.

The shortest path between the selected cities is highlighted in red, helping users visually identify the optimal route. This graphical representation improves understanding by showing how different cities are connected and how the algorithm determines the shortest path.

CONCLUSION

The Shortest Path Finder project successfully demonstrates how graph data structures and Dijkstra Algorithm can be used to find the minimum distance between nodes. The system efficiently calculates the shortest path and provides accurate results for weighted graphs.

This project helps in understanding the practical application of data structures in real-world problems such as navigation and network routing. Overall, the system is simple, efficient, and easy to implement, making it a useful solution for path optimization problems.



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Thank You!!

